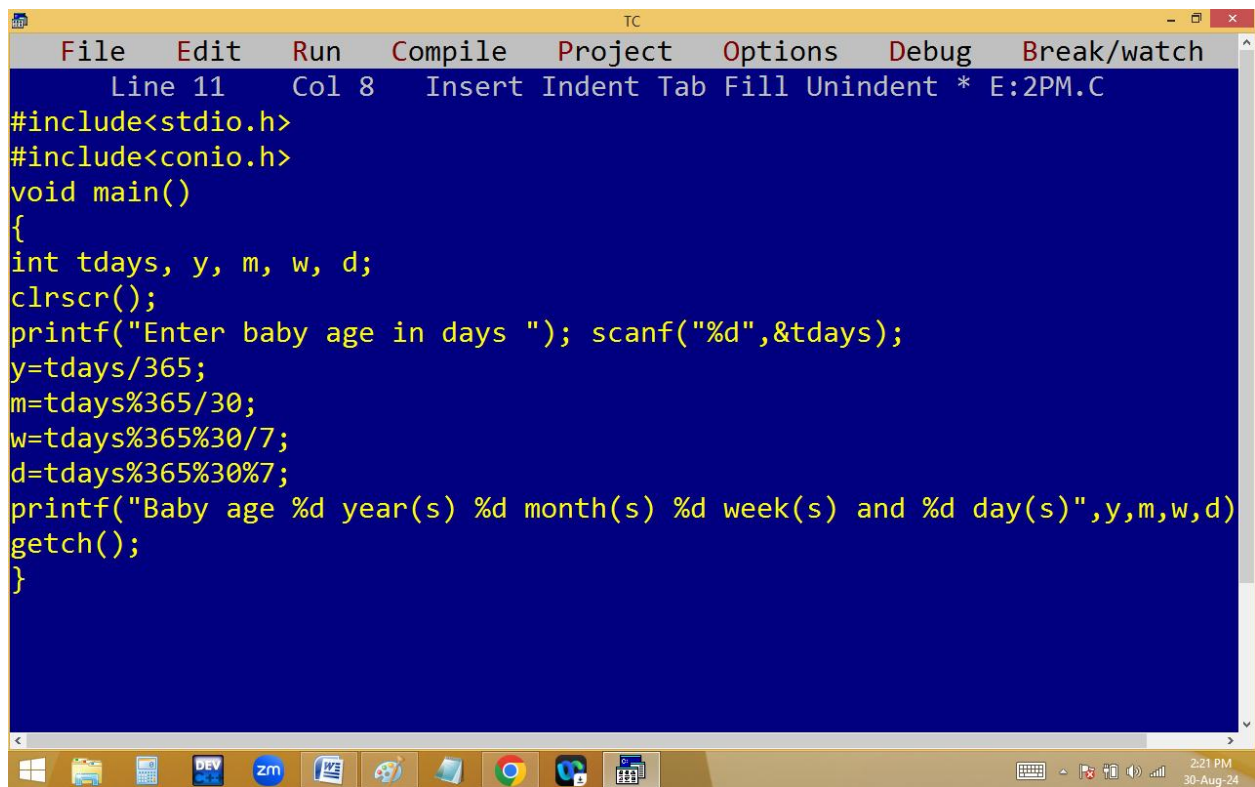
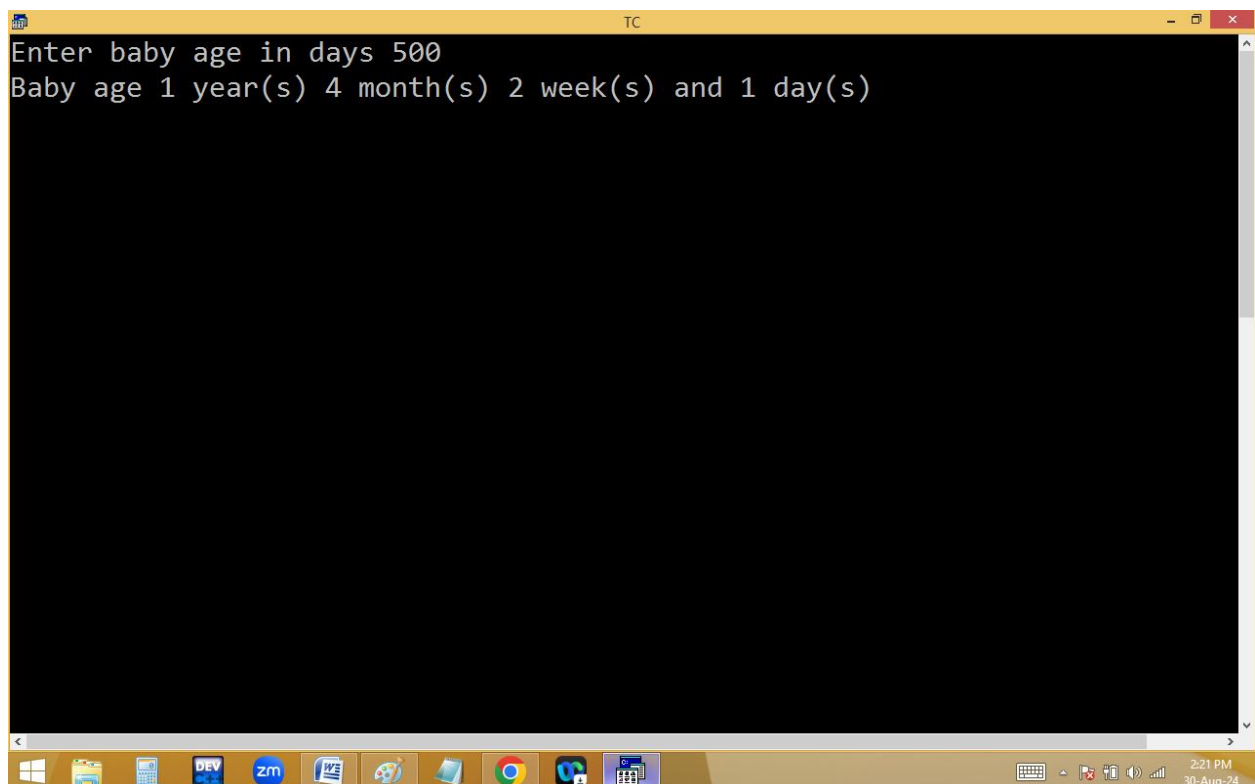


Finding baby age in years, months, weeks and days using total days.



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 8 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int tdays, y, m, w, d;
clrscr();
printf("Enter baby age in days "); scanf("%d",&tdays);
y=tdays/365;
m=tdays%365/30;
w=tdays%365%30/7;
d=tdays%365%30%7;
printf("Baby age %d year(s) %d month(s) %d week(s) and %d day(s)",y,m,w,d);
getch();
}
```



```
TC
Enter baby age in days 500
Baby age 1 year(s) 4 month(s) 2 week(s) and 1 day(s)
```

```
TC
Enter baby age in days 30000
Baby age 82 year(s) 2 month(s) 1 week(s) and 3 day(s)_
```

```
TC
Enter baby age in days 40000
Baby age -69 year(s) -11 month(s) -3 week(s) and 0 day(s)_
```

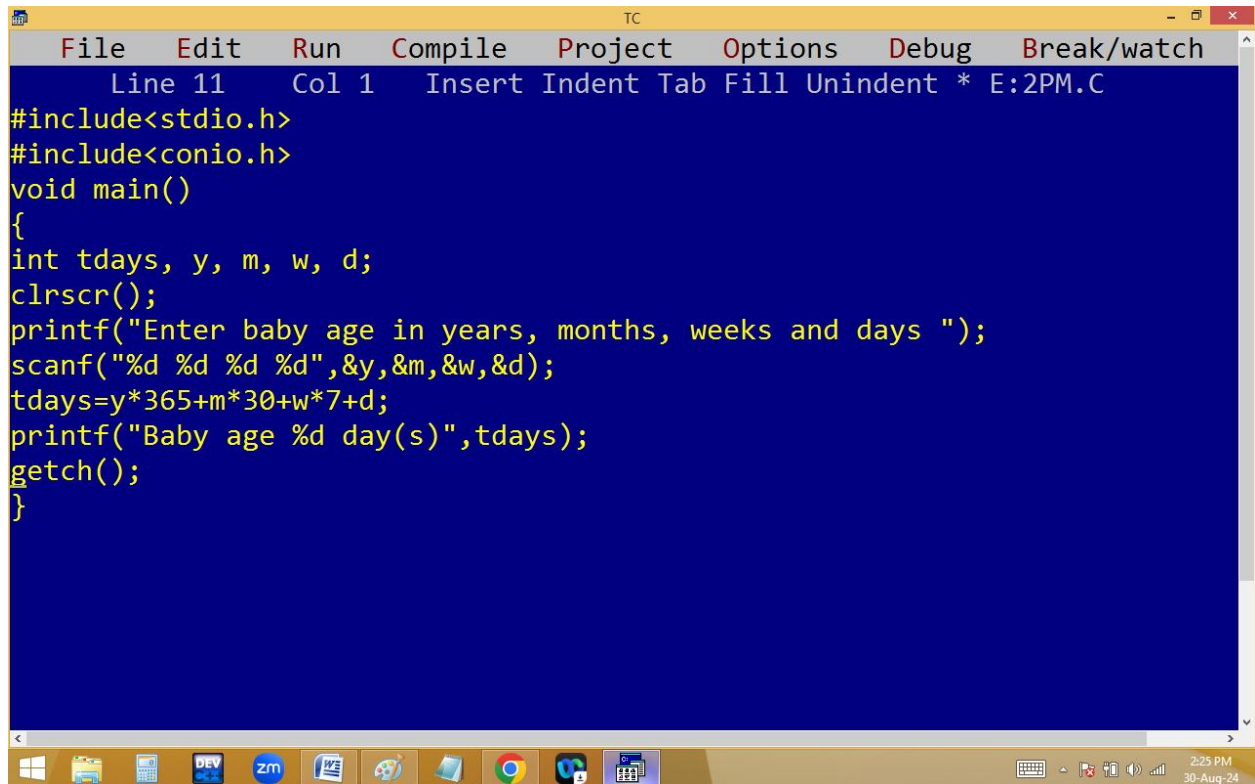
```
TC
Enter baby age in days 200
Baby age 0 year(s) 6 month(s) 2 week(s) and 6 day(s)
```

```
TC
Enter baby age in days -3
Baby age 0 year(s) 0 month(s) 0 week(s) and -3 day(s)_
```

Read baby age in years, months, weeks and days. Find baby age in days.

$$1 \text{ y} + 4 \text{ m} + 2 \text{ w} + 1 \text{ d} = 500 \text{ days}$$

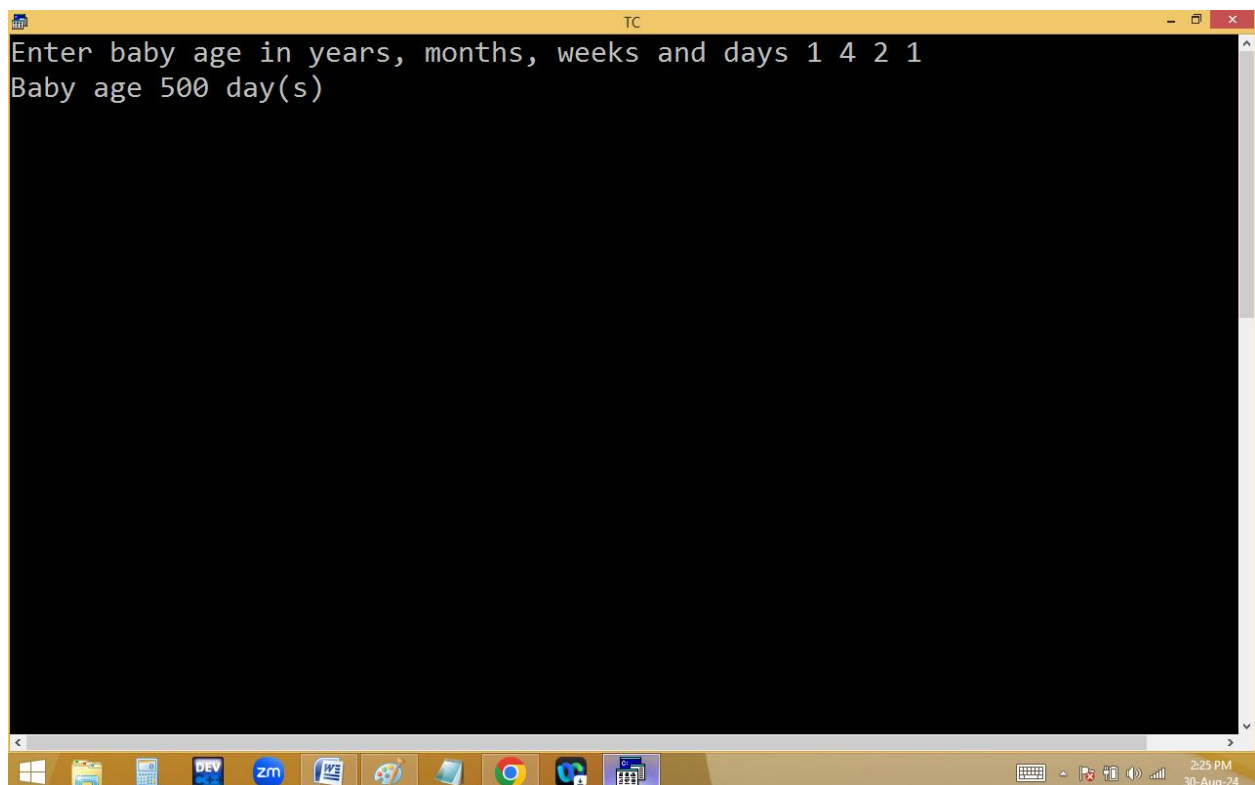
$$1 * 365 + 4 * 30 + 2 * 7 + 1 = 500$$



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 11 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int tdays, y, m, w, d;
clrscr();
printf("Enter baby age in years, months, weeks and days ");
scanf("%d %d %d %d",&y,&m,&w,&d);
tdays=y*365+m*30+w*7+d;
printf("Baby age %d day(s)",tdays);
getch();
}
```

The Windows taskbar at the bottom shows icons for Windows, File Explorer, Calculator, DEV C++, Zm, Word, Paint, and Google Chrome. The system clock in the bottom right corner displays '2:25 PM 30-Aug-24'.



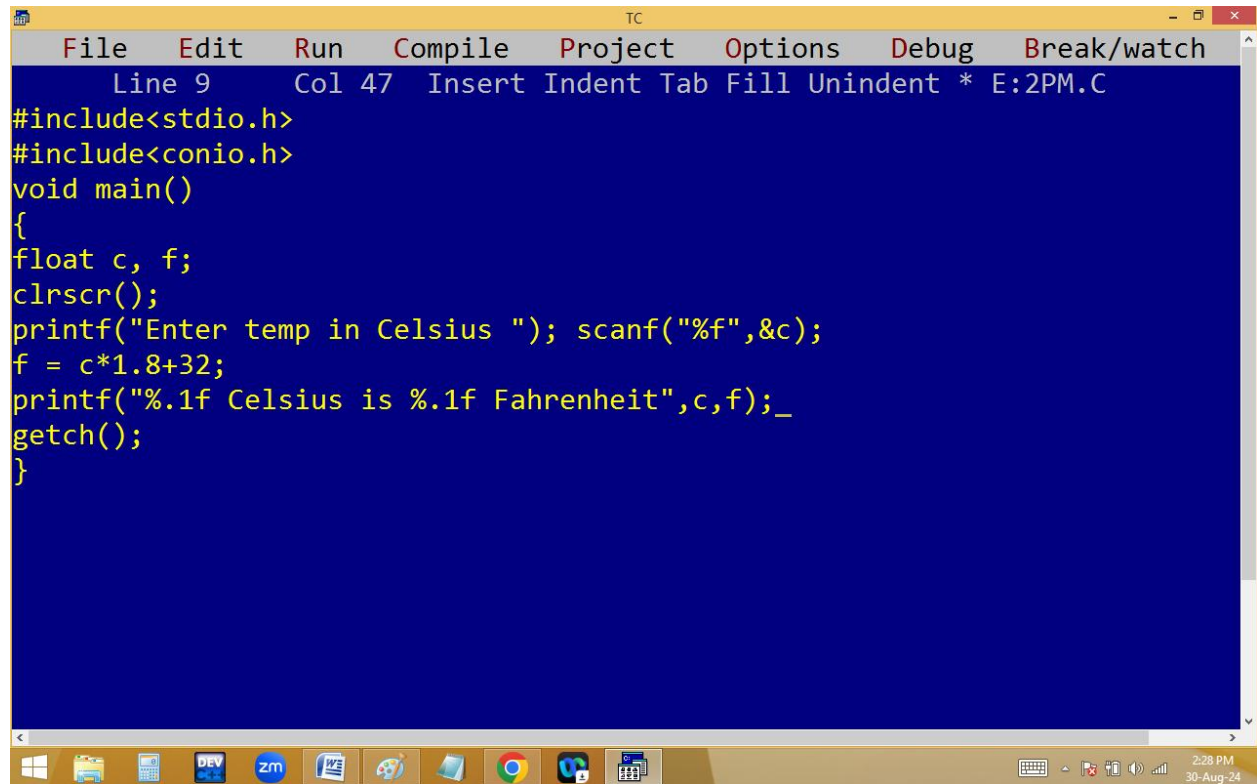
The screenshot shows the Turbo C++ (TC) IDE with a black background, displaying the output of the program. The text shown is:

```
Enter baby age in years, months, weeks and days 1 4 2 1
Baby age 500 day(s)
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the same set of application icons and system clock information ('2:25 PM 30-Aug-24').

Celsius to Fahrenheit conversion:

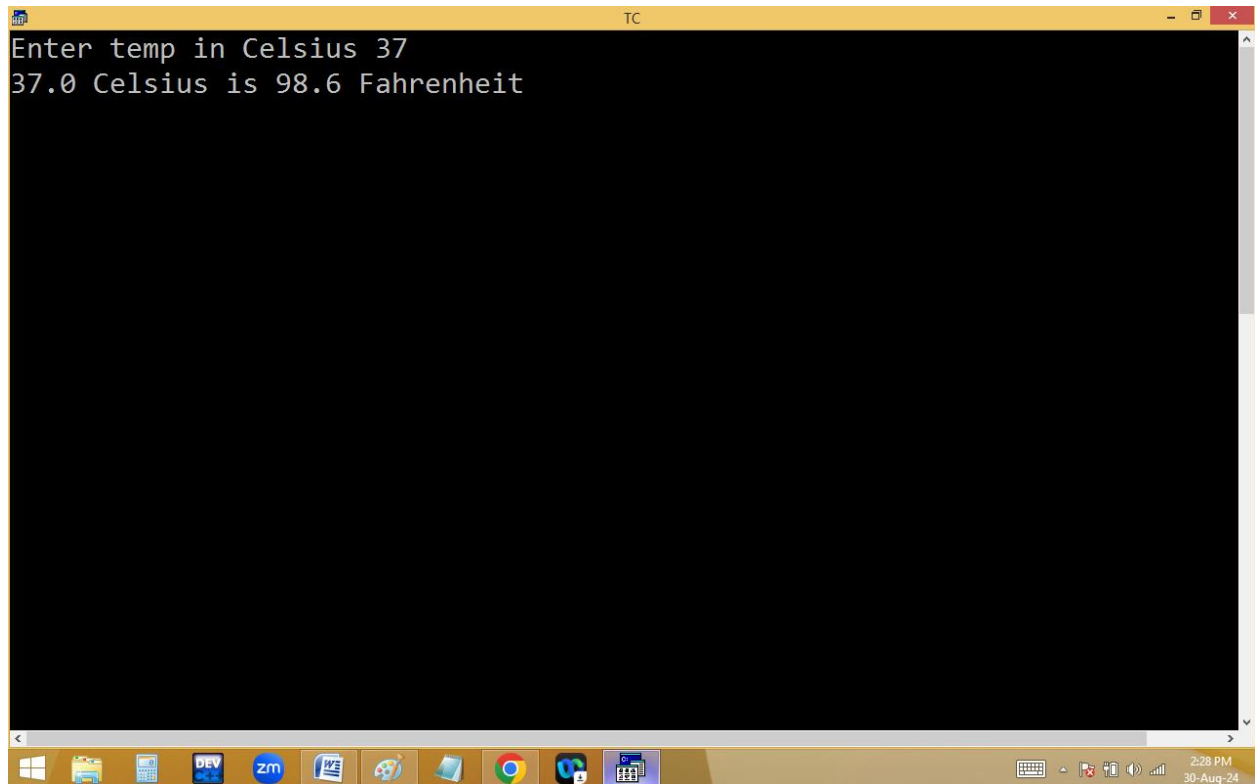
$$F = c * 1.8 + 32$$



The screenshot shows a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 9", "Col 47", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a dark blue background with yellow text. The code is as follows:

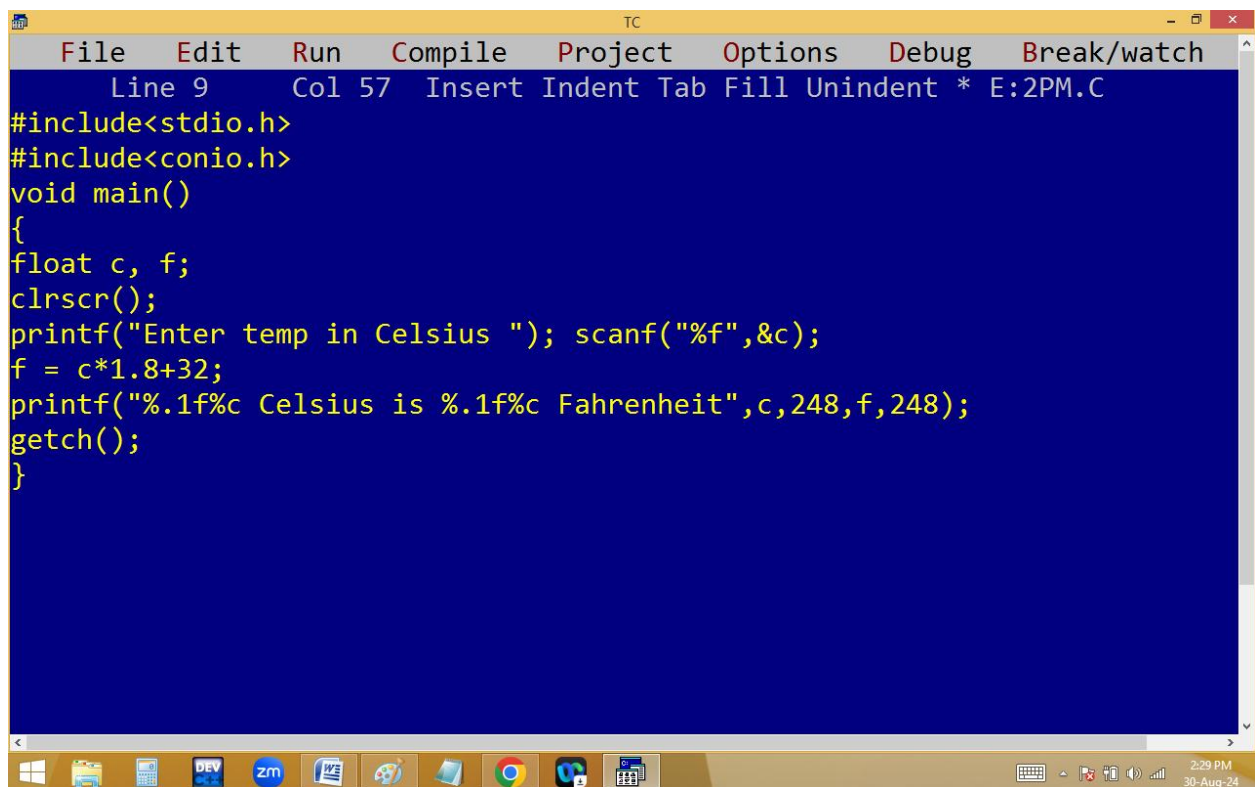
```
#include<stdio.h>
#include<conio.h>
void main()
{
float c, f;
clrscr();
printf("Enter temp in Celsius "); scanf("%f",&c);
f = c*1.8+32;
printf("%.1f Celsius is %.1f Fahrenheit",c,f);_
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, a calculator, DEV C++, Zoom, a folder, a document, Google Chrome, and a task manager icon. The system clock in the bottom right corner shows "2:28 PM" and "30-Aug-24".



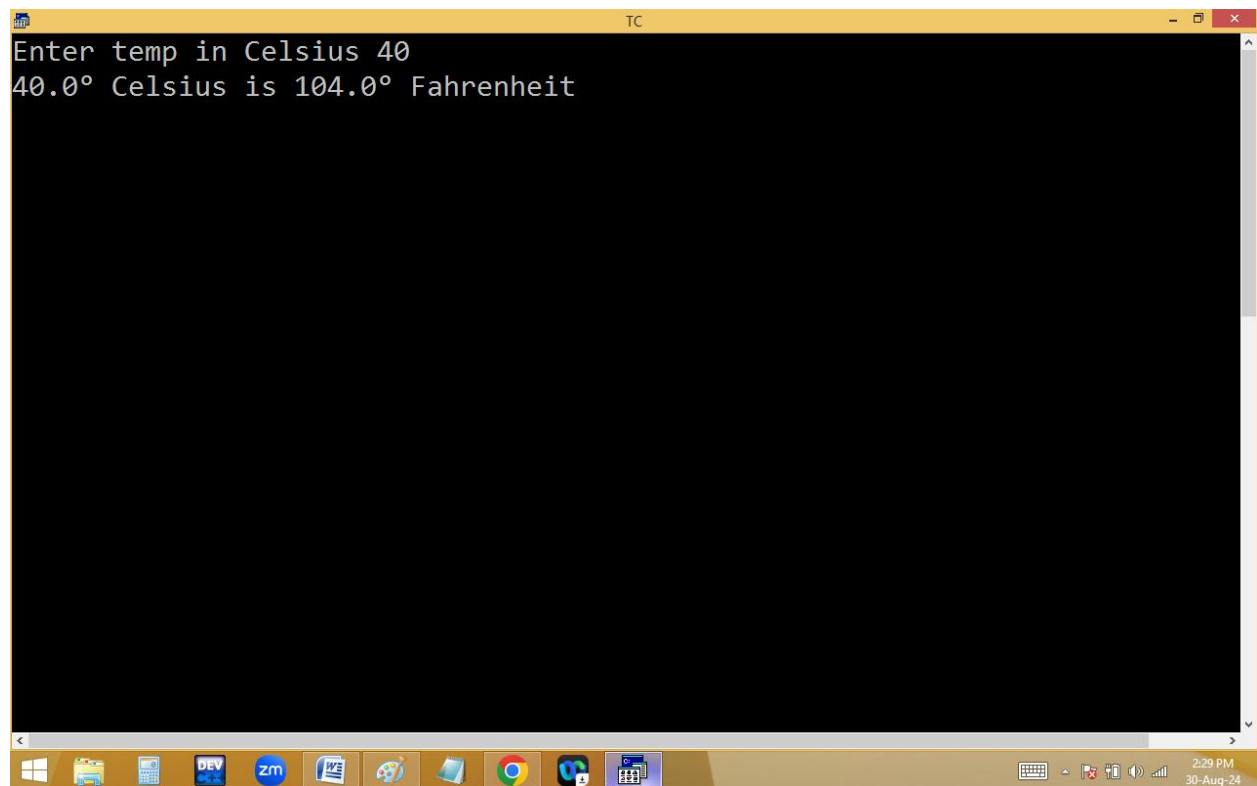
A screenshot of a Turbo C++ (TC) console window. The window has a yellow title bar with the text "TC" and standard minimize, maximize, and close buttons. The main area is black with white text. The text shows the program's execution: "Enter temp in Celsius 37" followed by "37.0 Celsius is 98.6 Fahrenheit". The Windows taskbar is visible at the bottom, showing icons for various applications and the system clock indicating 2:28 PM on 30-Aug-24.

```
Enter temp in Celsius 37
37.0 Celsius is 98.6 Fahrenheit
```



A screenshot of a Turbo C++ (TC) editor window. The window has a yellow title bar with "TC" and standard window controls. Below the title bar is a menu bar with "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A status bar at the top of the editor area shows "Line 9", "Col 57", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editor area has a blue background with yellow text showing the source code for a program that converts Celsius to Fahrenheit. The code includes headers for stdio.h and conio.h, defines a main function, declares float variables c and f, clears the screen, prompts for input, calculates the Fahrenheit value, and prints the result. The Windows taskbar is visible at the bottom, showing icons for various applications and the system clock indicating 2:29 PM on 30-Aug-24.

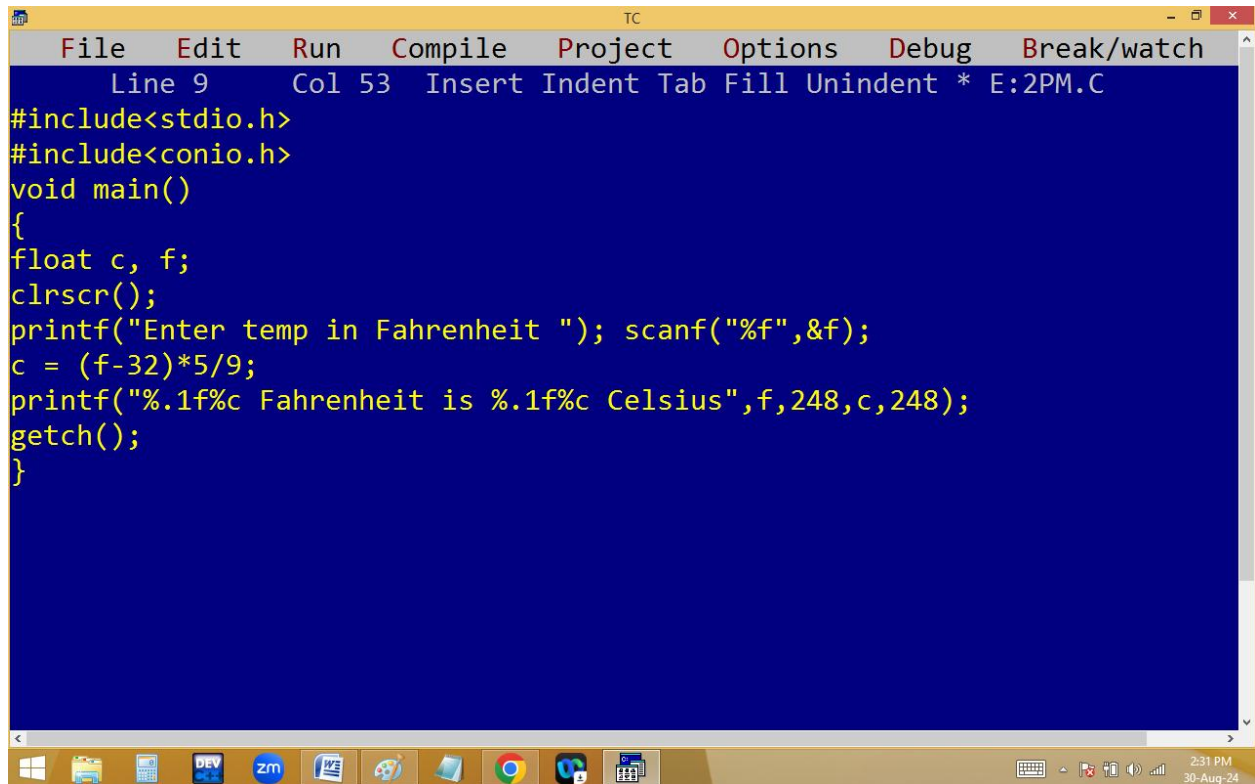
```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 57 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
float c, f;
clrscr();
printf("Enter temp in Celsius "); scanf("%f",&c);
f = c*1.8+32;
printf("%.1f%c Celsius is %.1f%c Fahrenheit",c,248,f,248);
getch();
}
```



```
TC
Enter temp in Celsius 40
40.0° Celsius is 104.0° Fahrenheit
```

Fahrenheit to Celsius:

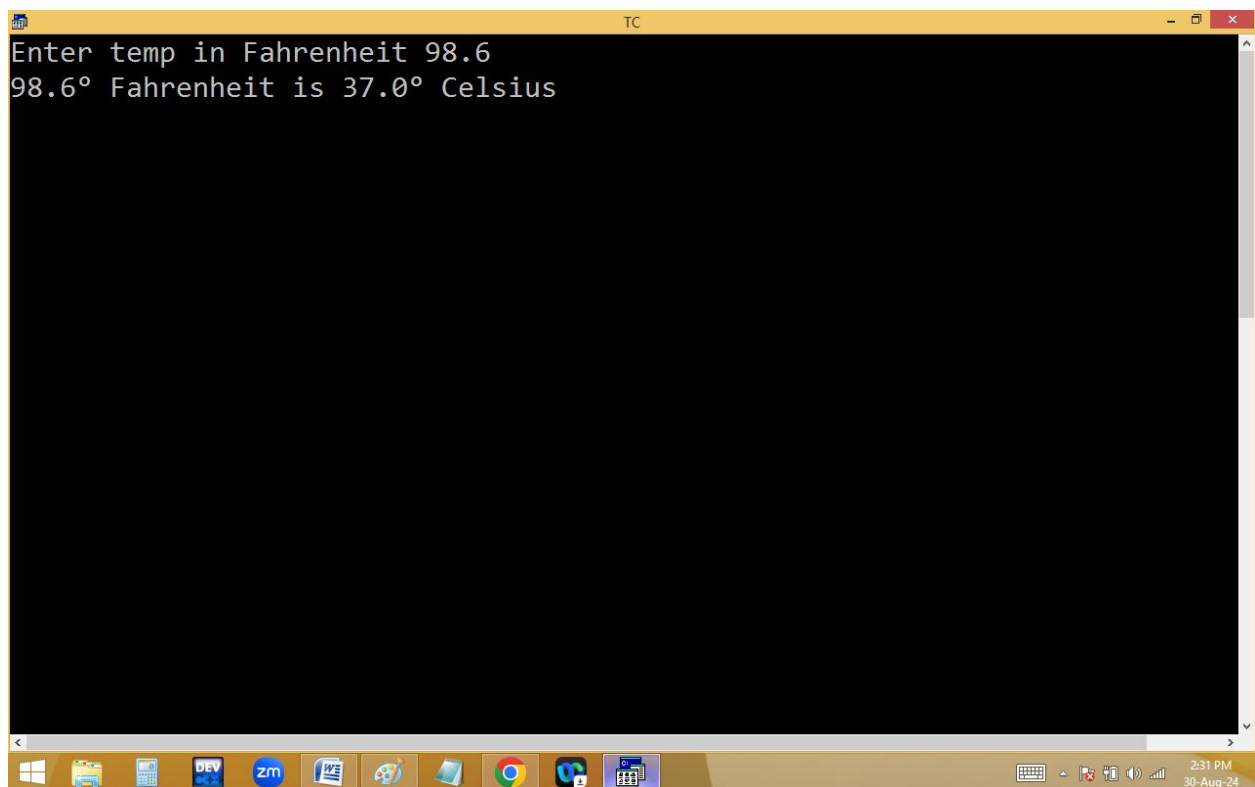
$$C = (f-32)*5/9;$$



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 9 Col 53 Insert Indent Tab Fill Unindent * E:2PM.C'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float c, f;
clrscr();
printf("Enter temp in Fahrenheit "); scanf("%f",&f);
c = (f-32)*5/9;
printf("%.1f° Fahrenheit is %.1f° Celsius",f,248,c,248);
getch();
}
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others. The system clock in the bottom right corner displays '2:31 PM 30-Aug-24'.



The screenshot shows the Turbo C++ (TC) IDE with a black background, displaying the output of the program. The text shown is:

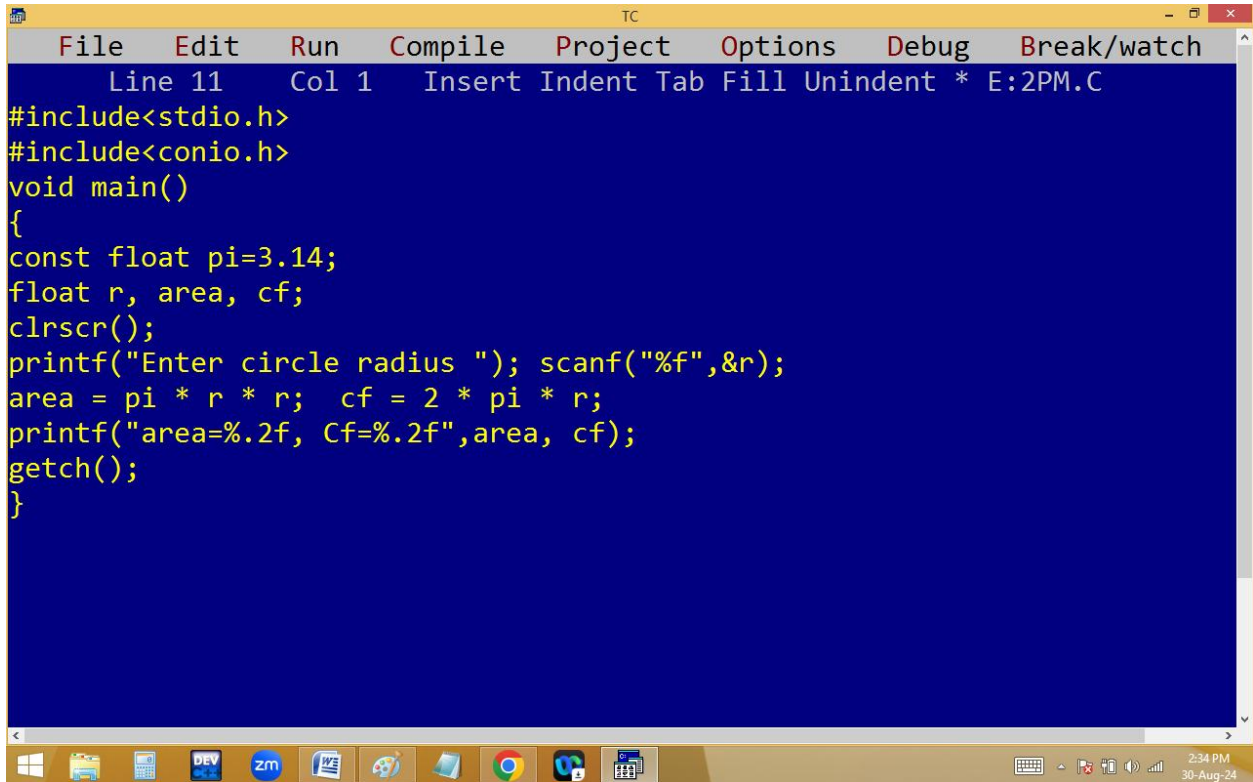
```
Enter temp in Fahrenheit 98.6
98.6° Fahrenheit is 37.0° Celsius
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the same icons and system clock ('2:31 PM 30-Aug-24').

Finding area and circumference of a circle.

$\text{Area} = \pi * r * r$

$\text{Cf} = 2 * \pi * r$

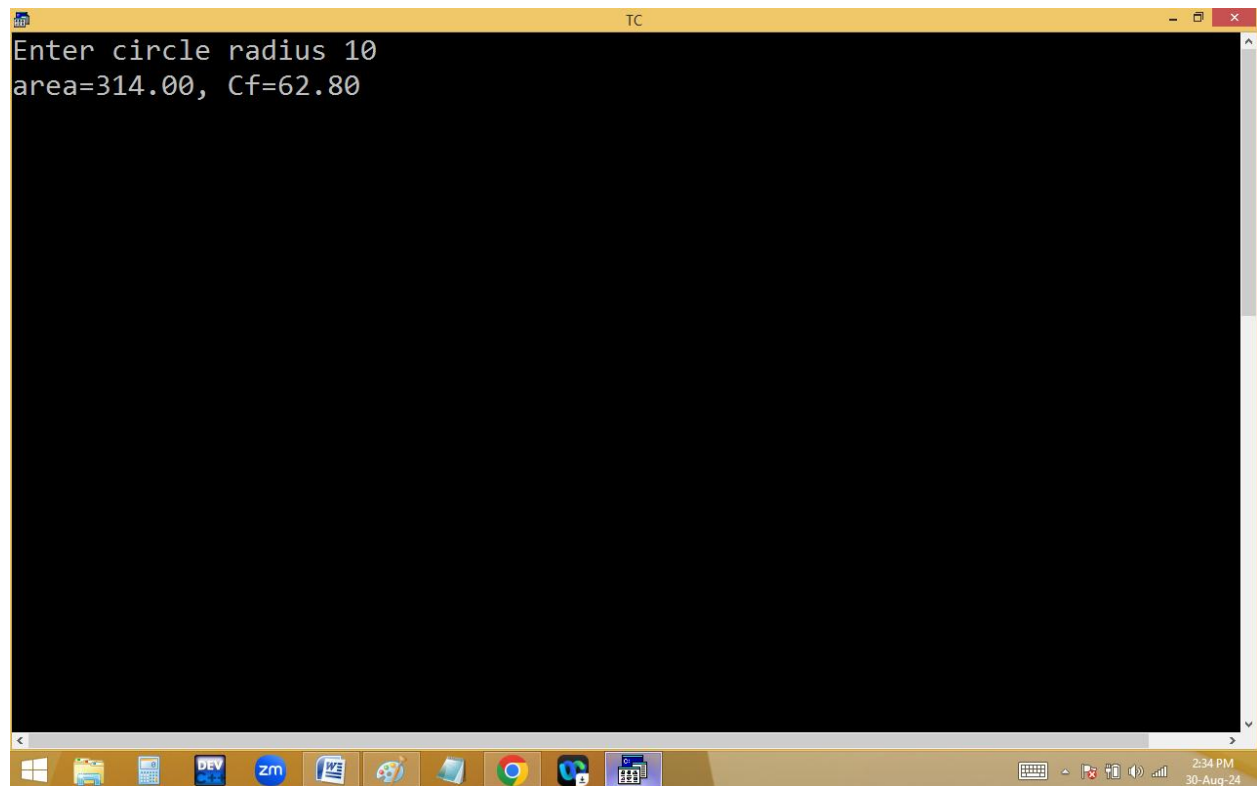


The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 11 Col 1' and 'Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

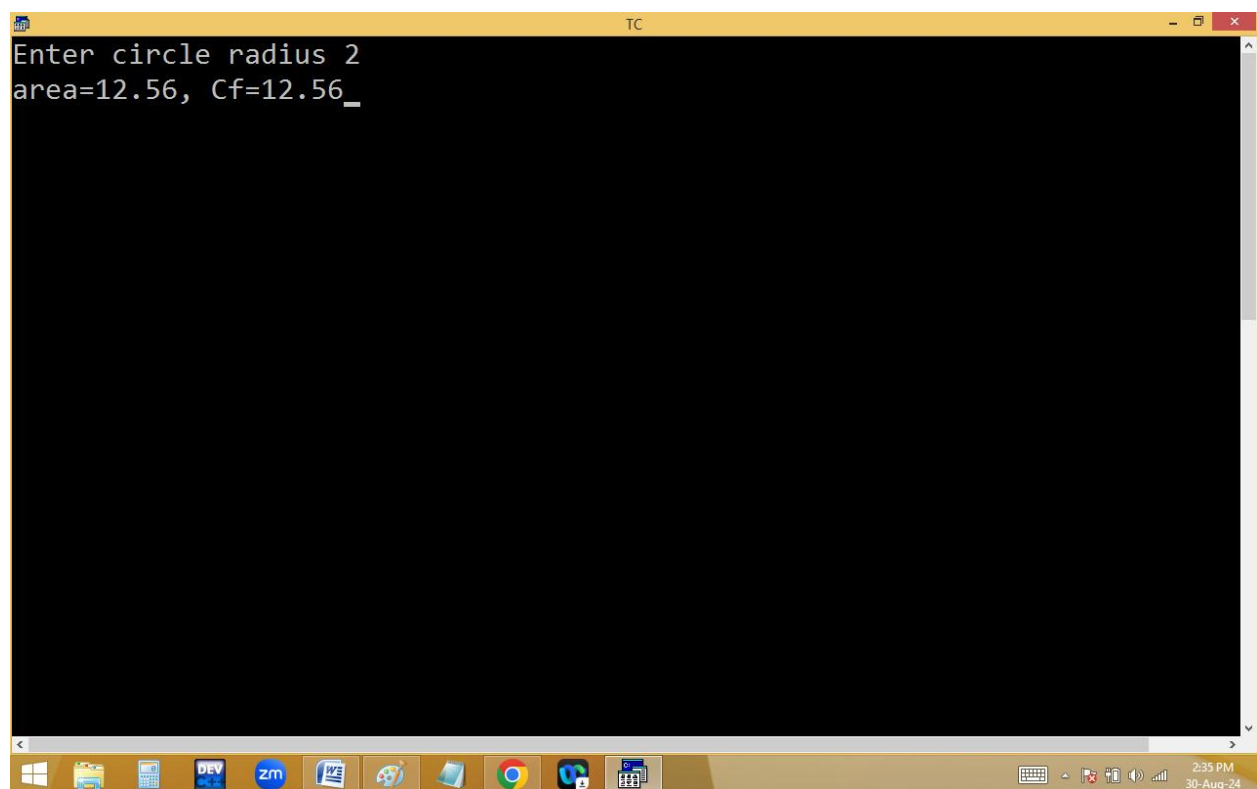
```
#include<stdio.h>
#include<conio.h>
void main()
{
    const float pi=3.14;
    float r, area, cf;
    clrscr();
    printf("Enter circle radius "); scanf("%f",&r);
    area = pi * r * r;  cf = 2 * pi * r;
    printf("area=%.2f, Cf=%.2f",area, cf);
    getch();
}
```

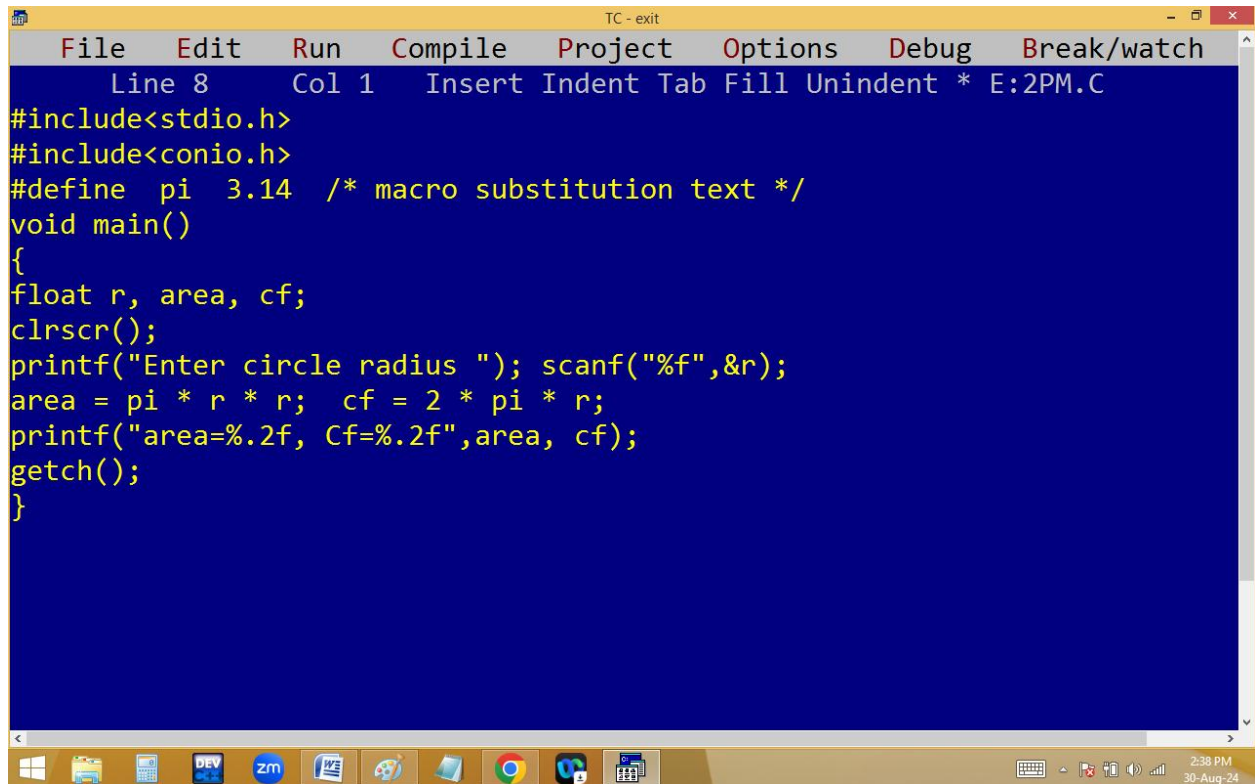
The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, zm, Word, Paint, Chrome, and a folder icon. The system clock in the bottom right corner shows '2:34 PM' and '30-Aug-24'.

```
TC
Enter circle radius 10
area=314.00, Cf=62.80
```



```
TC
Enter circle radius 2
area=12.56, Cf=12.56_
```

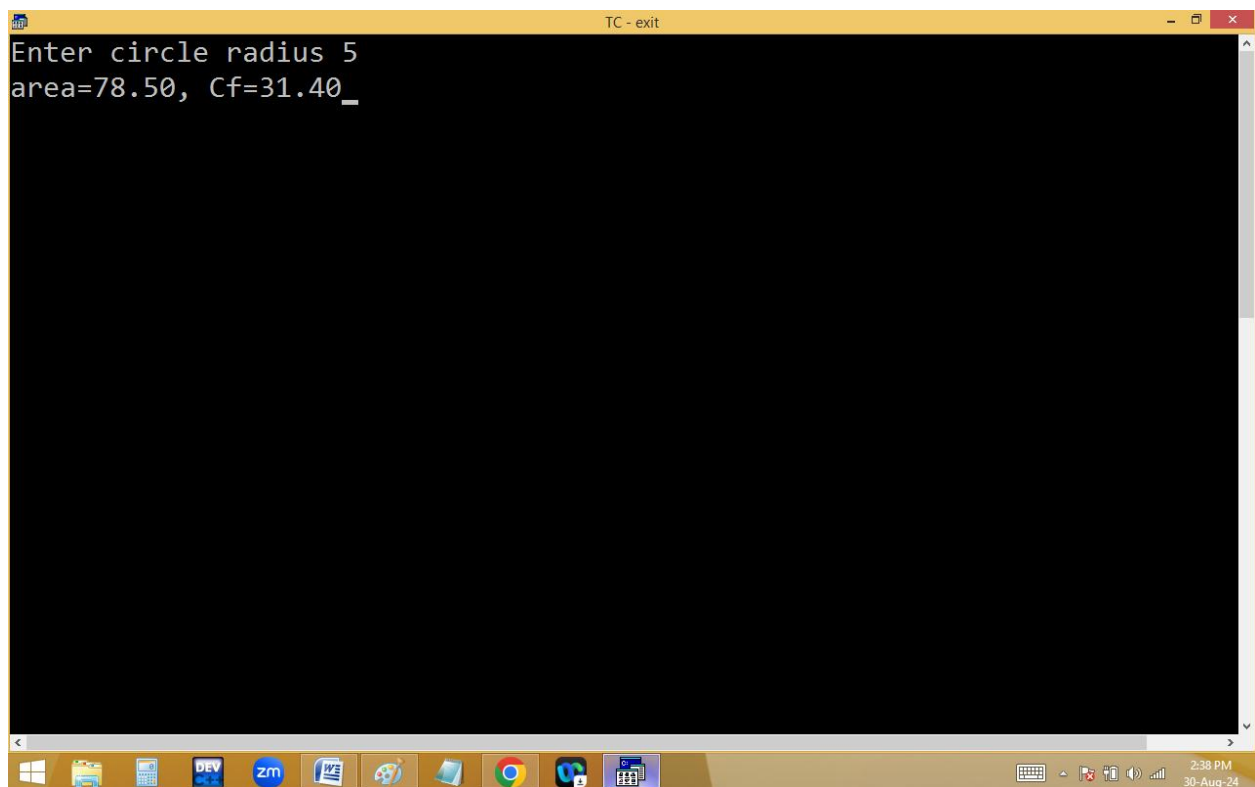




The screenshot shows the Turbo C++ IDE with the following code in the editor:

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#define pi 3.14 /* macro substitution text */
void main()
{
float r, area, cf;
clrscr();
printf("Enter circle radius "); scanf("%f",&r);
area = pi * r * r; cf = 2 * pi * r;
printf("area=%.2f, Cf=%.2f",area, cf);
getch();
}
```

The Windows taskbar at the bottom shows the time as 2:38 PM on 30-Aug-24.



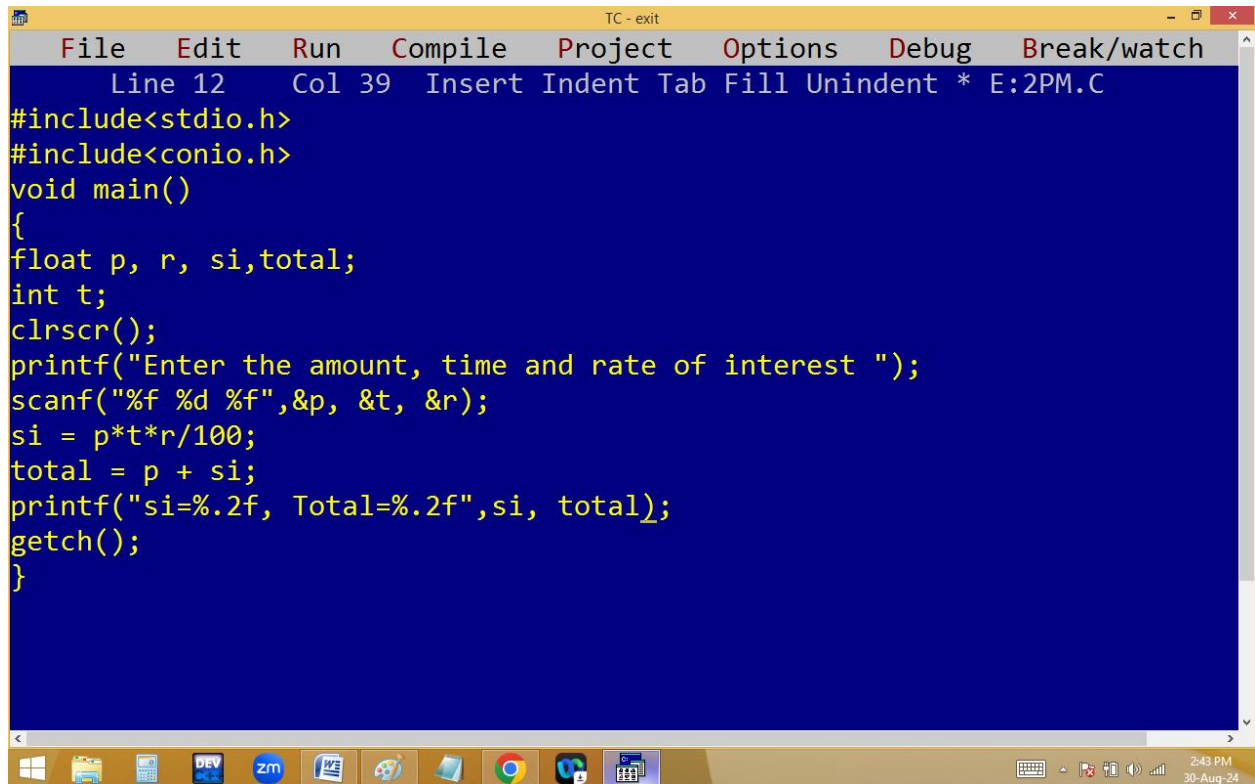
The screenshot shows the Turbo C++ IDE with the following output in the console:

```
Enter circle radius 5
area=78.50, Cf=31.40_
```

The Windows taskbar at the bottom shows the time as 2:38 PM on 30-Aug-24.

Finding simple interest:

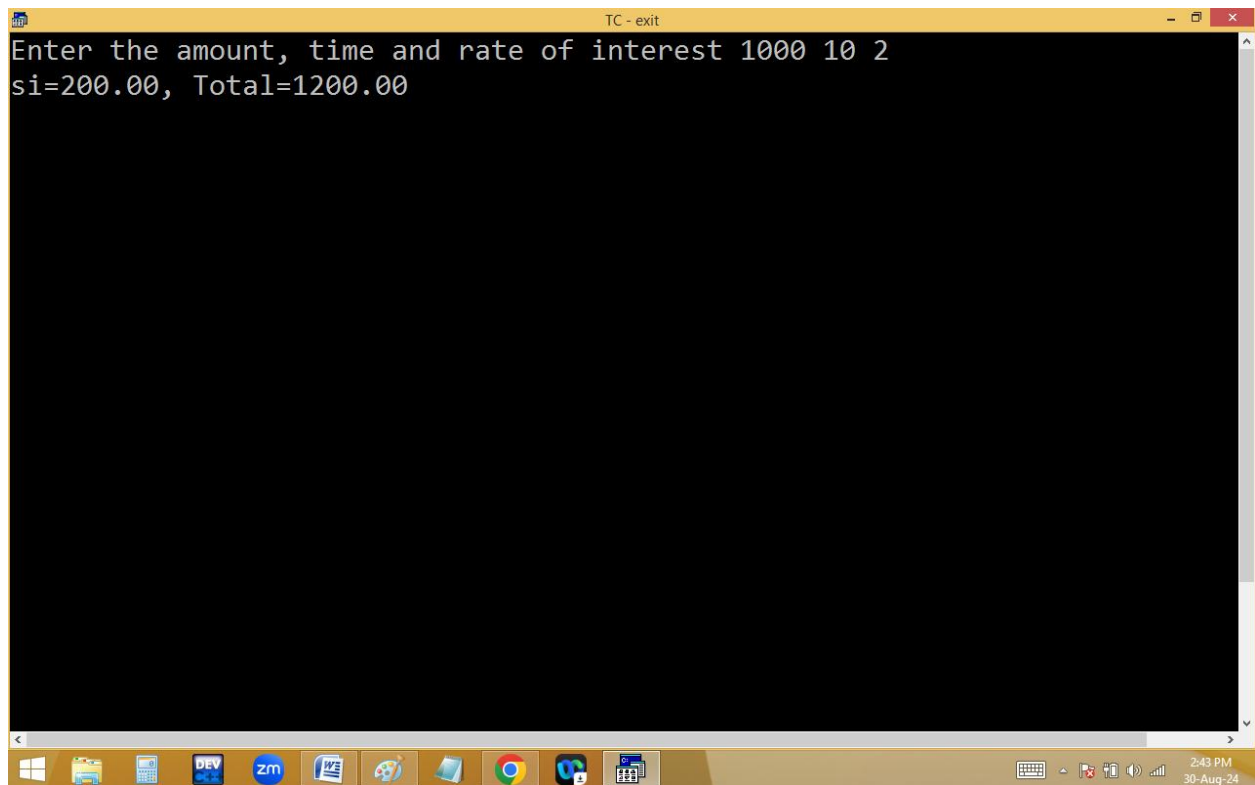
$$\begin{array}{rcl} 2 \times 10 = 20 & \leftarrow & \underline{1000} - P \\ \times & & 10 - t \\ \hline 10 & & \\ \hline si = 200 & & 2 - r \\ 1000 & & \\ \hline 1200 - \text{Total} & & \end{array}$$



The screenshot shows the Turbo C++ IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 12, Col 39, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float p, r, si,total;
int t;
clrscr();
printf("Enter the amount, time and rate of interest ");
scanf("%f %d %f",&p, &t, &r);
si = p*t*r/100;
total = p + si;
printf("si=%.2f, Total=%.2f",si, total);
getch();
}
```

The Windows taskbar at the bottom shows icons for Windows, File Explorer, DEV, zm, Word, Paint, Chrome, and a calendar. The system clock indicates 2:43 PM on 30-Aug-24.

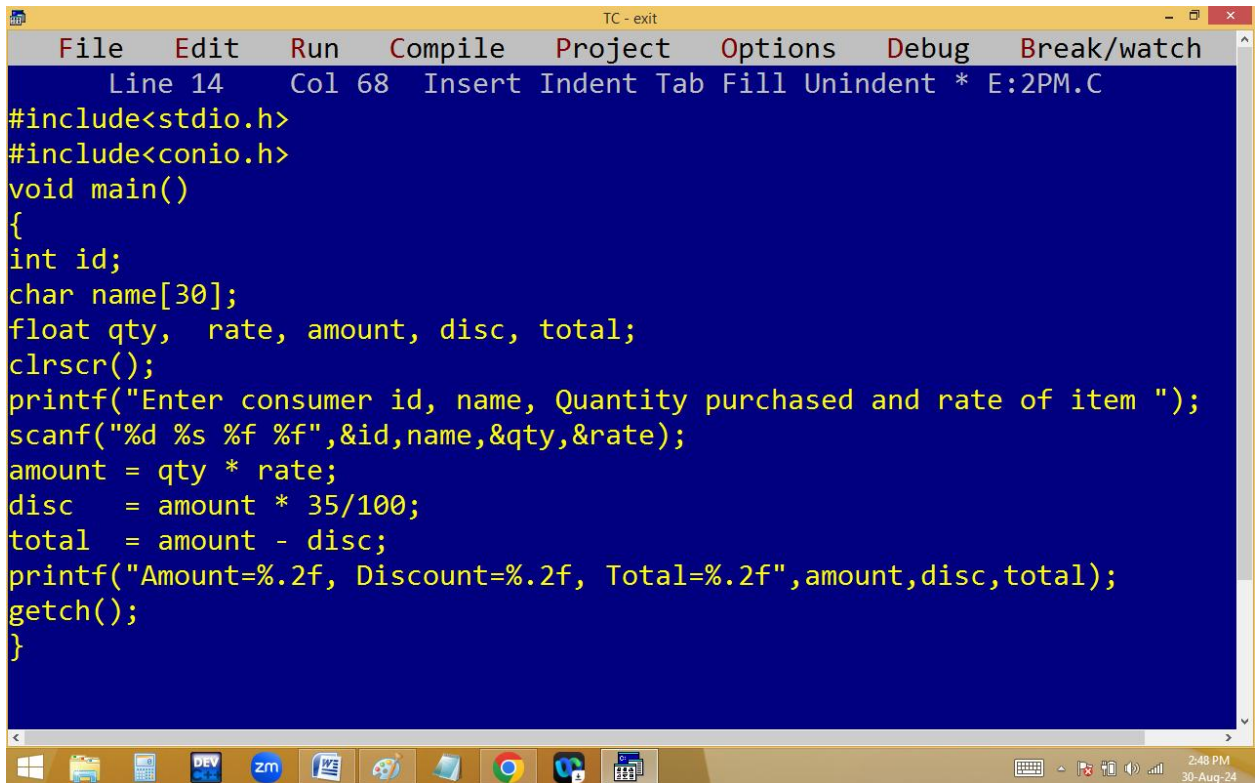


The screenshot shows the Turbo C++ IDE with the same menu bar and status bar. The output window displays the following text:

```
Enter the amount, time and rate of interest 1000 10 2
si=200.00, Total=1200.00
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the same icons and system clock (2:43 PM on 30-Aug-24).

Read a customer id, name, Quantity purchased and rate of item. Find the amount, 35% discount and total.



The image shows a screenshot of a Turbo C++ IDE window titled "TC - exit". The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates "Line 14 Col 68 Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a dark blue background with yellow text showing the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int id;
char name[30];
float qty, rate, amount, disc, total;
clrscr();
printf("Enter consumer id, name, Quantity purchased and rate of item ");
scanf("%d %s %f %f",&id,name,&qty,&rate);
amount = qty * rate;
disc   = amount * 35/100;
total  = amount - disc;
printf("Amount=%.2f, Discount=%.2f, Total=%.2f",amount,disc,total);
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for various applications and the system clock indicating 2:48 PM on 30-Aug-24.

```
TC - exit
Enter consumer id, name, Quantity purchased and rate of item
1001 Krish 10 100
Amount=1000.00, Discount=350.00, Total=650.00
```

```
TC - exit
Enter consumer id, name, Quantity purchased and rate of item
1 kumari 1 1
Amount=1.00, Discount=0.35, Total=0.65_
```

```
TC - exit
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 26 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int id;
char name[30];
float qty, rate, amount, disc, total;
clrscr();
printf("Enter consumer id, name, Quantity purchased and rate of item ");
scanf("%d %s %f %f",&id,name,&qty,&rate);
amount = qty * rate;
printf("Amount=%.2f\n",amount);
printf("Enter discount percentage ");scanf("%f",&disc);
disc = amount * disc/100;
total = amount - disc;
printf("Amount=%.2f, Discount=%.2f, Total=%.2f",amount,disc,total);
getch();
}
```

```
TC - exit
Enter consumer id, name, Quantity purchased and rate of item
1 kumari 1 1
Amount=1.00
Enter discount percentage 0
Amount=1.00, Discount=0.00, Total=1.00
```

```
TC - exit
Enter consumer id, name, Quantity purchased and rate of item
2 bablu 1 1000
Amount=1000.00
Enter discount percentage 20
Amount=1000.00, Discount=200.00, Total=800.00_
```

```
TC - exit
Enter consumer id, name, Quantity purchased and rate of item
3 wife 1 10000
Amount=10000.00
Enter discount percentage 100
Amount=10000.00, Discount=10000.00, Total=0.00_
```

CONTROL STATEMENTS / CONTROL STRUCTURES

They are used to control program execution flow [order]. In c we can control program execution by using below statements.

CONTROL STATEMENTS / CONTROL STRUCTURES

<u>unconditional branching</u>	<u>Conditional Branching</u>	<u>Loops/Iterations</u>
1. goto label	1. if	1. entry / pre controlled
2. break	a. simple if	a. while
3. continue	b. if..else	b. for
4. return	c. if..else if ladder	2. exit / post controlled
5. exit() // function	d. nested if	a. do..while
	2. switch	

goto label / jumping statement

It is used to transfer program execution from one place to another place [label].

In this process it is jumping from one area to another without any condition. Hence it

is also called **unconditional** jumping statement.

Syntax:



```
.....;  
.....;  
goto label;  
.....;  
.....;  
label:  
.....;  
.....;
```

Here **goto** is a keyword.

Label is an identifier is used to identify the area[line].

Every label should be end with : (colon)

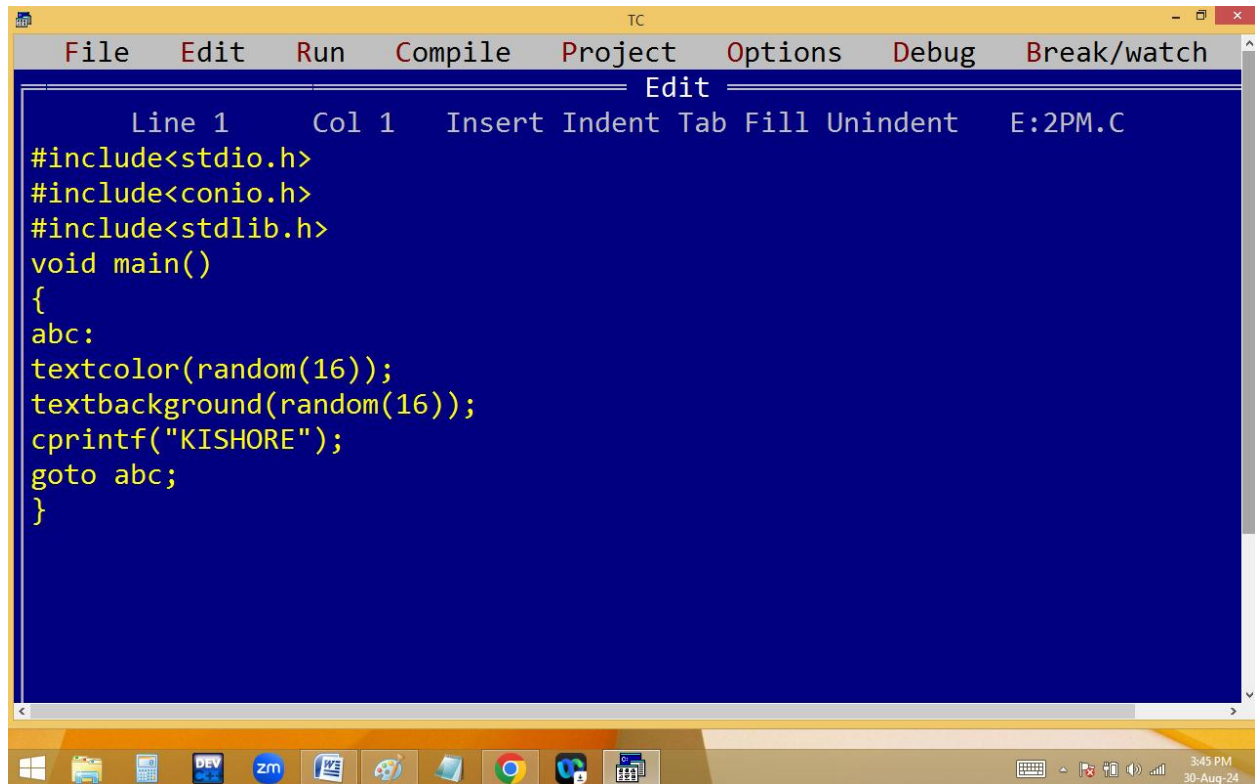
Keywords not allowed in labels i.e. label should be user defined.

Duplicate labels not allowed.

There is no space between go and to.

Label naming rules are similar to the identifier rules.

Note: goto label working style is similar to loops some times.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The code editor contains the following C program:

```
Line 1 Col 1 Insert Indent Tab Fill Unindent E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
abc:
textcolor(random(16));
textbackground(random(16));
cprintf("KISHORE");
goto abc;
}
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others. The system clock indicates 3:45 PM on 30-Aug-24.

